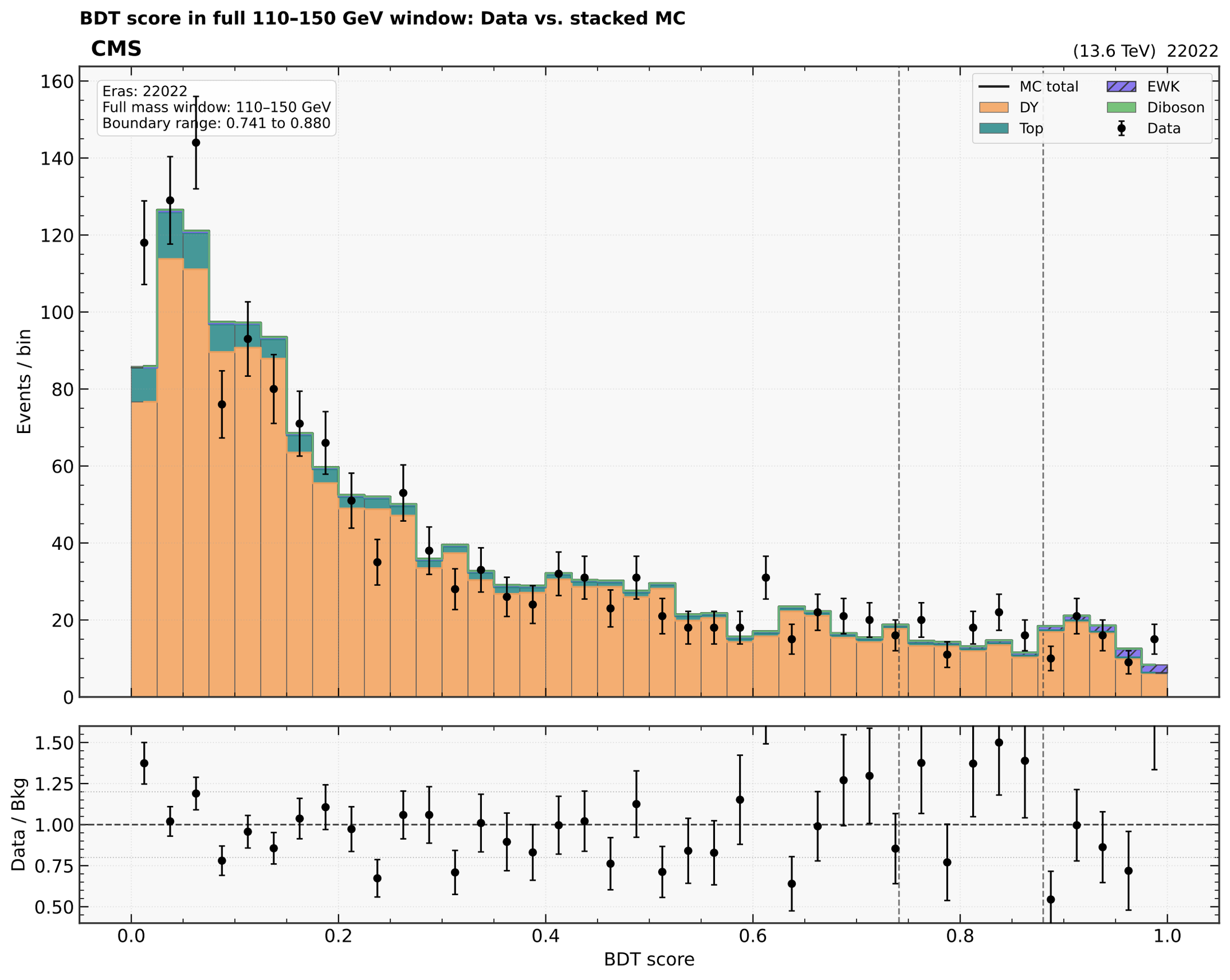
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 22022
Category edges : 0.0000, 0.7410, 0.8801, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ²/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_CB vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

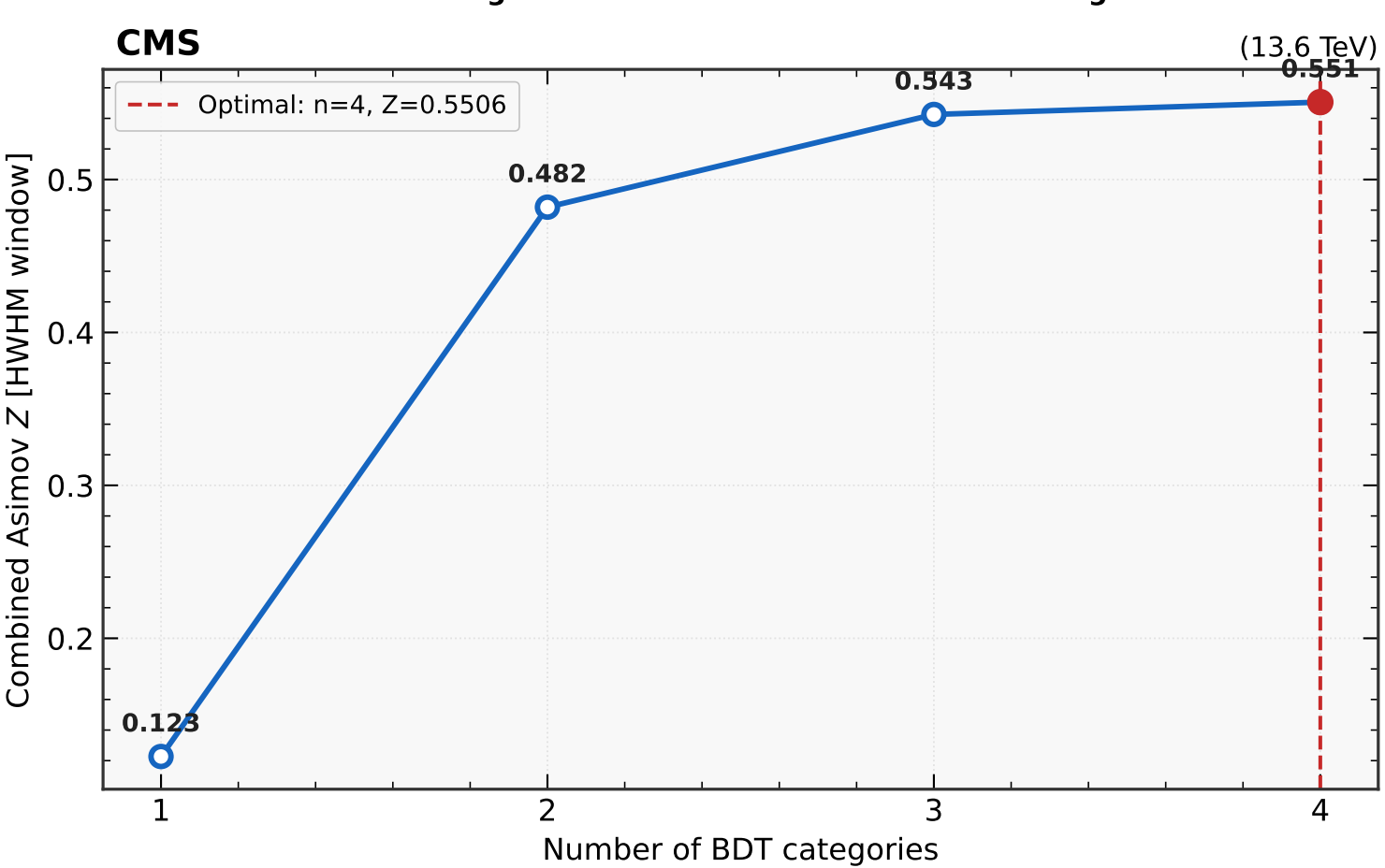

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 22022
Eras: 22022
Full mass window: 110–150 GeV
Boundary range: 0.741 to 0.880
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

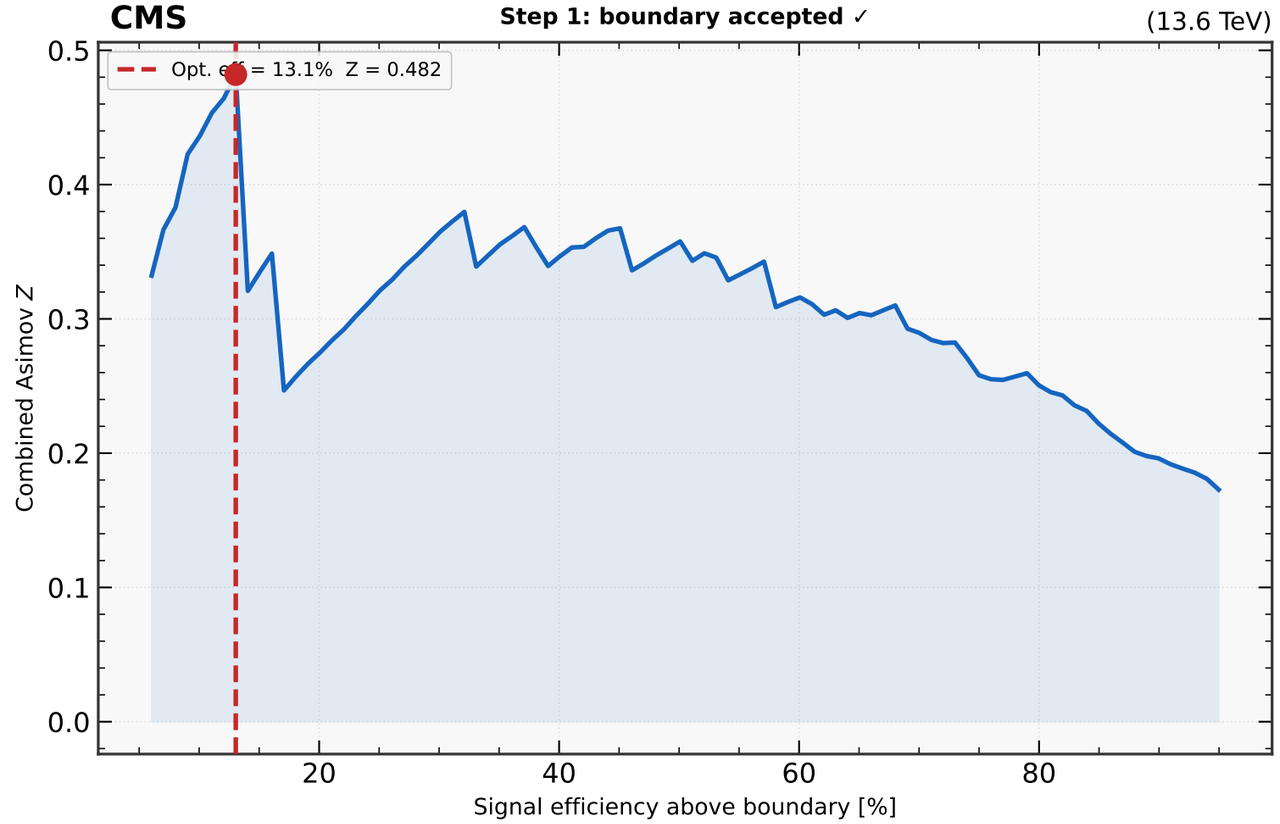
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.003	85.90 ± 6.15	118	0.00
[0.025, 0.050)	0.010	126.51 ± 7.46	129	0.01
[0.050, 0.075)	0.014	121.08 ± 7.62	144	0.01
[0.075, 0.100)	0.017	97.39 ± 6.82	76	0.02
[0.100, 0.125)	0.019	97.21 ± 7.01	93	0.02
[0.125, 0.150)	0.021	93.46 ± 6.75	80	0.02
[0.150, 0.175)	0.023	68.48 ± 5.62	71	0.03
[0.175, 0.200)	0.023	59.66 ± 5.34	66	0.04
[0.200, 0.225)	0.026	52.44 ± 5.01	51	0.05
[0.225, 0.250)	0.025	52.00 ± 5.06	35	0.05
[0.250, 0.275)	0.028	50.06 ± 4.68	53	0.06
[0.275, 0.300)	0.028	35.88 ± 4.01	38	0.08
[0.300, 0.325)	0.030	39.49 ± 4.22	28	0.08
[0.325, 0.350)	0.031	32.69 ± 3.78	33	0.09
[0.350, 0.375)	0.033	29.05 ± 3.64	26	0.11
[0.375, 0.400)	0.033	28.89 ± 3.62	24	0.12
[0.400, 0.425)	0.035	32.11 ± 3.86	32	0.11
[0.425, 0.450)	0.038	30.37 ± 3.80	31	0.12
[0.450, 0.475)	0.039	30.18 ± 4.00	23	0.13
[0.475, 0.500)	0.040	27.56 ± 3.66	31	0.14

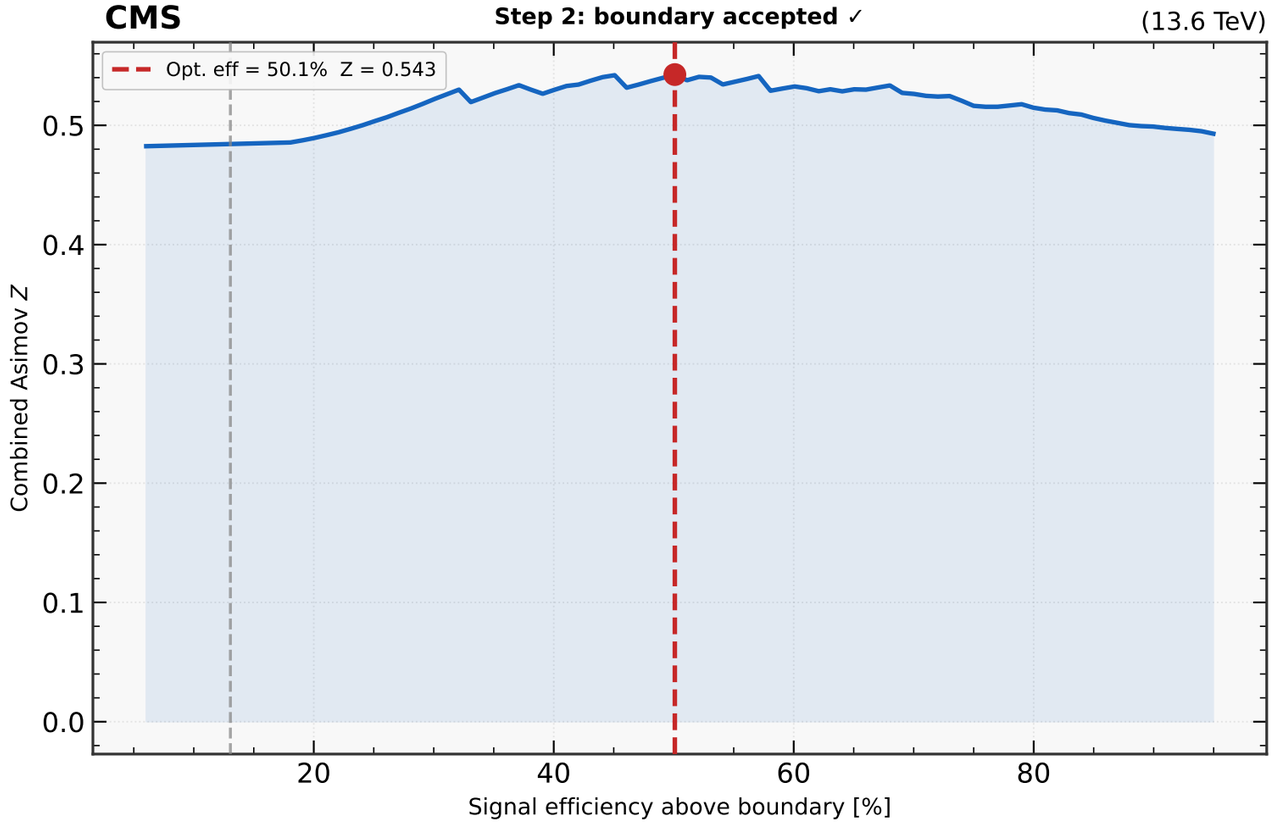
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

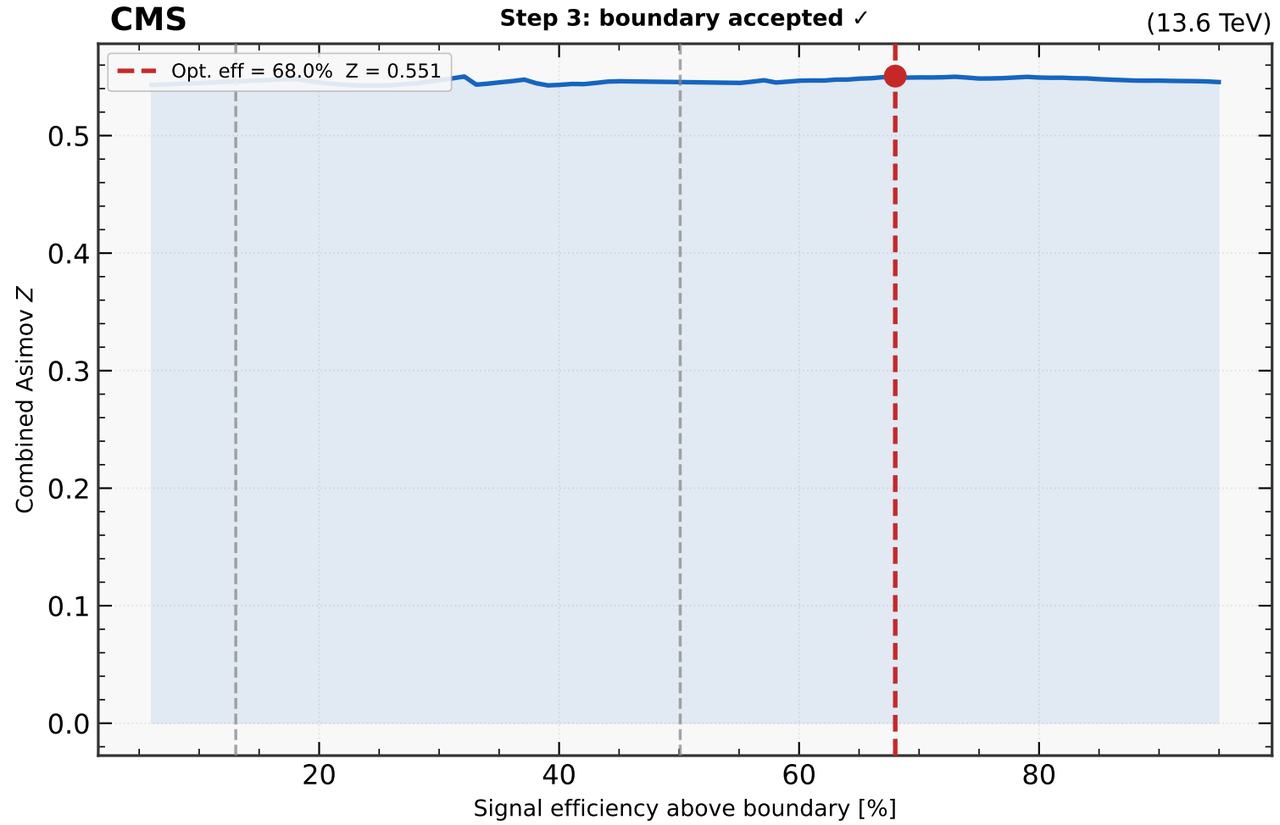
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.044	29.50 ± 3.72	21	0.15
[0.525, 0.550)	0.043	21.42 ± 2.84	18	0.20
[0.550, 0.575)	0.048	21.73 ± 3.13	18	0.22
[0.575, 0.600)	0.049	15.63 ± 2.57	18	0.31
[0.600, 0.625)	0.052	17.05 ± 2.76	31	0.30
[0.625, 0.650)	0.055	23.45 ± 3.32	15	0.23
[0.650, 0.675)	0.059	22.22 ± 3.32	22	0.26
[0.675, 0.700)	0.063	16.53 ± 2.74	21	0.38
[0.700, 0.725)	0.071	15.42 ± 2.50	20	0.46
[0.725, 0.750)	0.075	18.74 ± 3.05	16	0.40
[0.750, 0.775)	0.082	14.54 ± 2.70	20	0.56
[0.775, 0.800)	0.091	14.28 ± 2.53	11	0.63
[0.800, 0.825)	0.102	13.12 ± 2.42	18	0.77
[0.825, 0.850)	0.118	14.67 ± 2.54	22	0.80
[0.850, 0.875)	0.133	11.52 ± 2.24	16	1.14
[0.875, 0.900)	0.166	18.38 ± 2.90	10	0.89
[0.900, 0.925)	0.207	21.08 ± 3.36	21	0.97
[0.925, 0.950)	0.281	18.55 ± 2.97	16	1.49
[0.950, 0.975)	0.437	12.52 ± 2.19	9	3.37
[0.975, 1.000)	0.581	8.34 ± 1.67	15	6.51


Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=0.5506
0.123
0.482
0.543
0.551
Combined Asimov Z [HWHM window]
0.2
0.3
0.4
0.5
1
2
3
4
Number of BDT categories

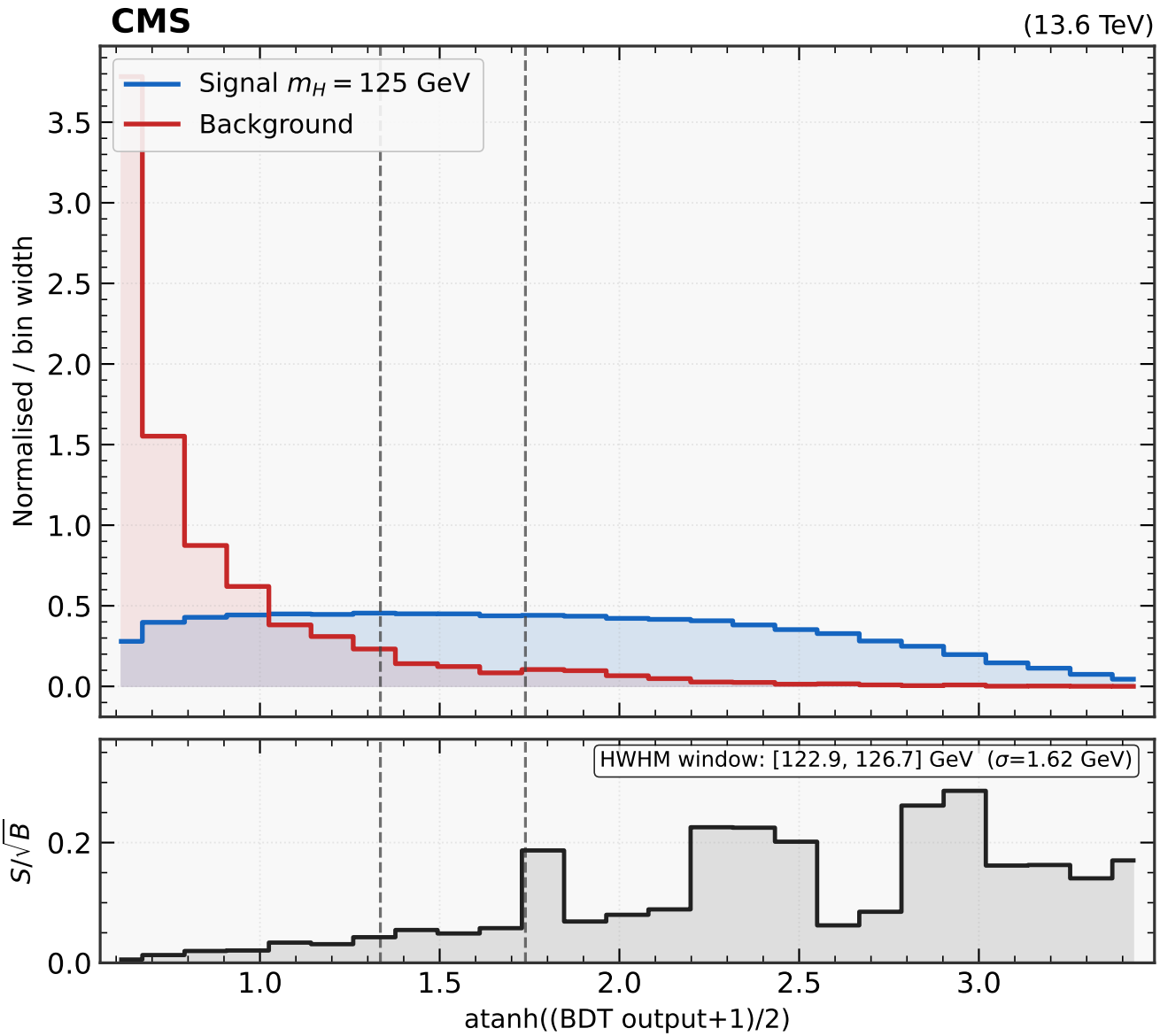
Iterative category boundary optimisation (AN Fig. 32 analog)

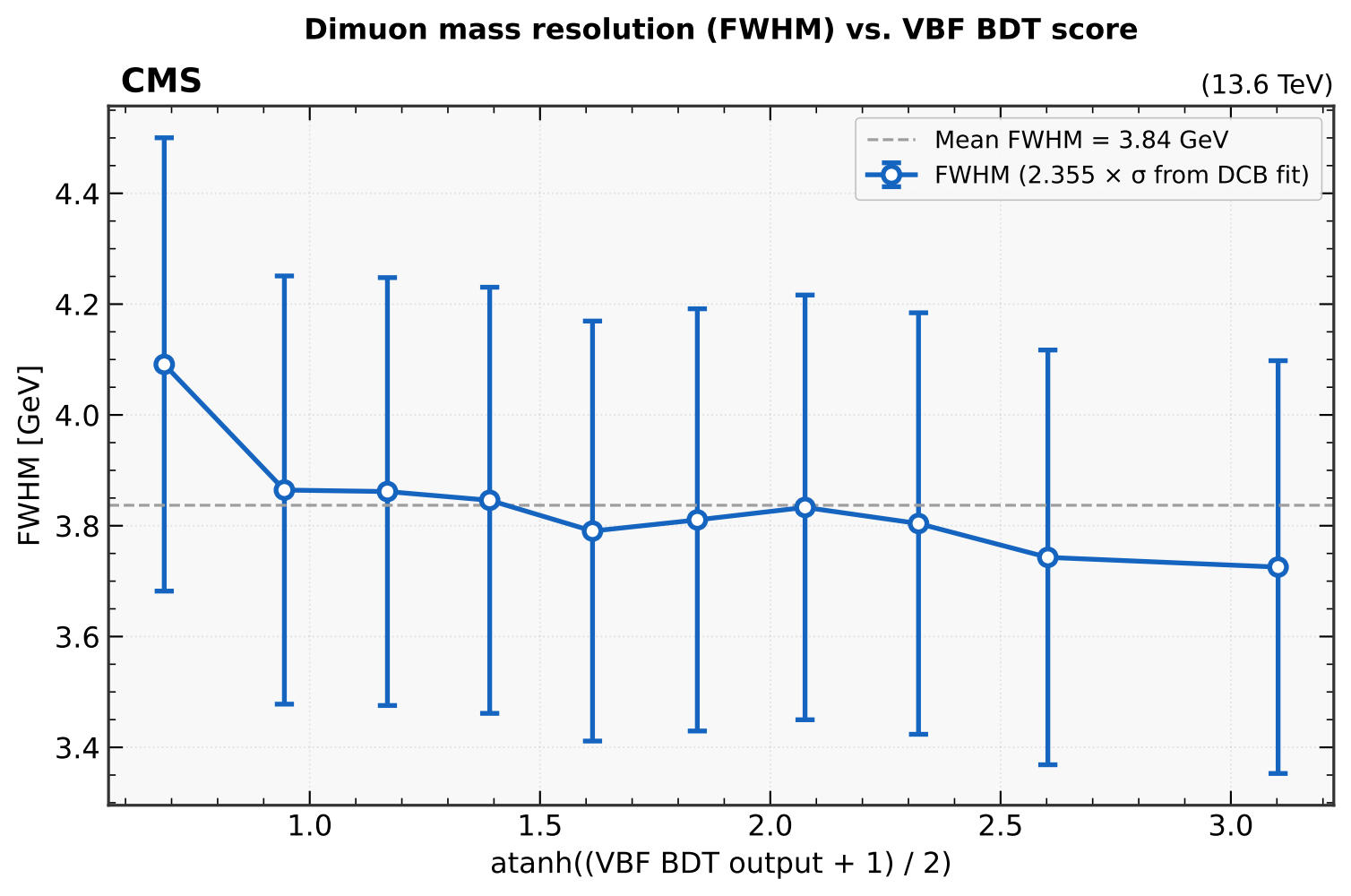


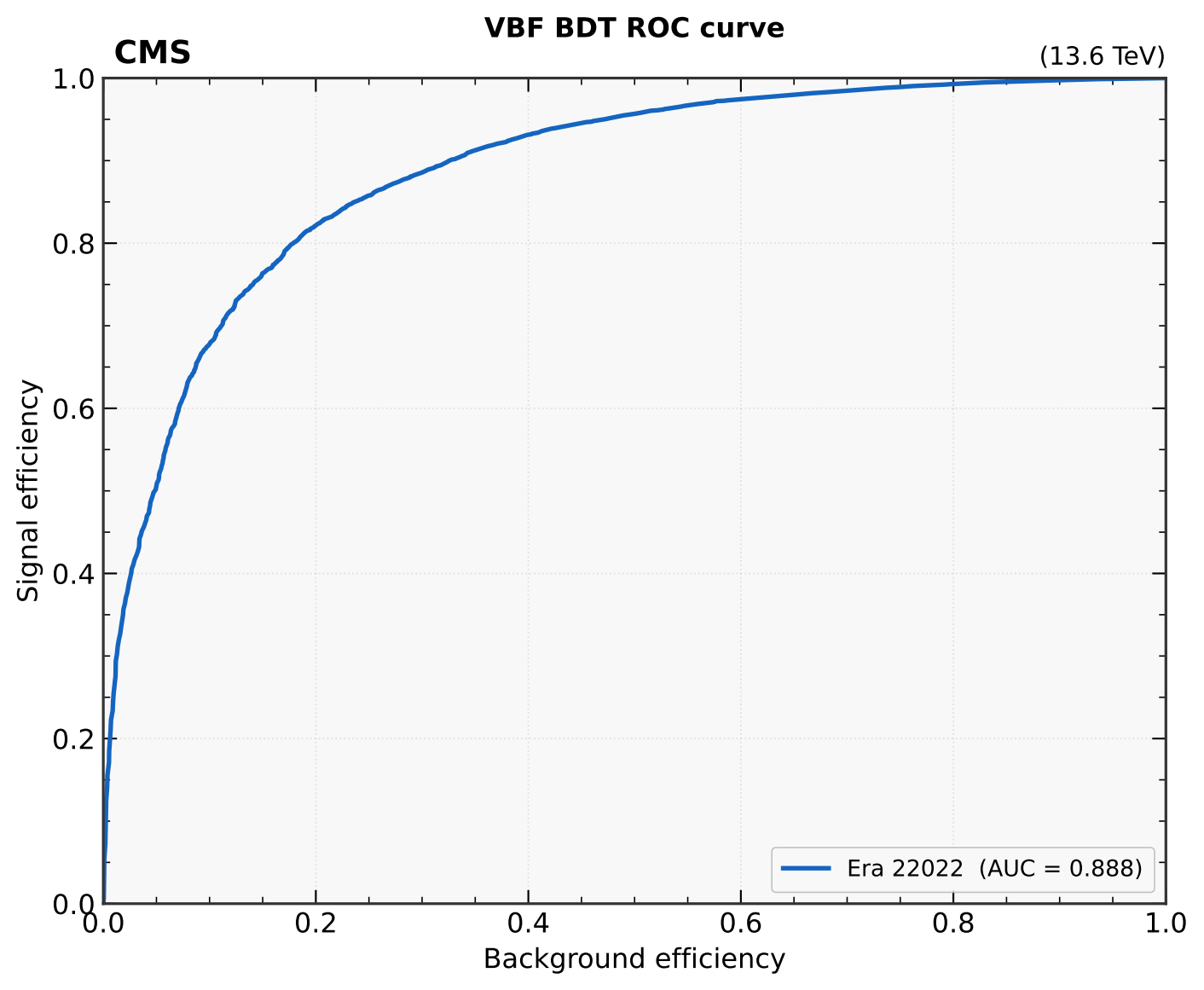


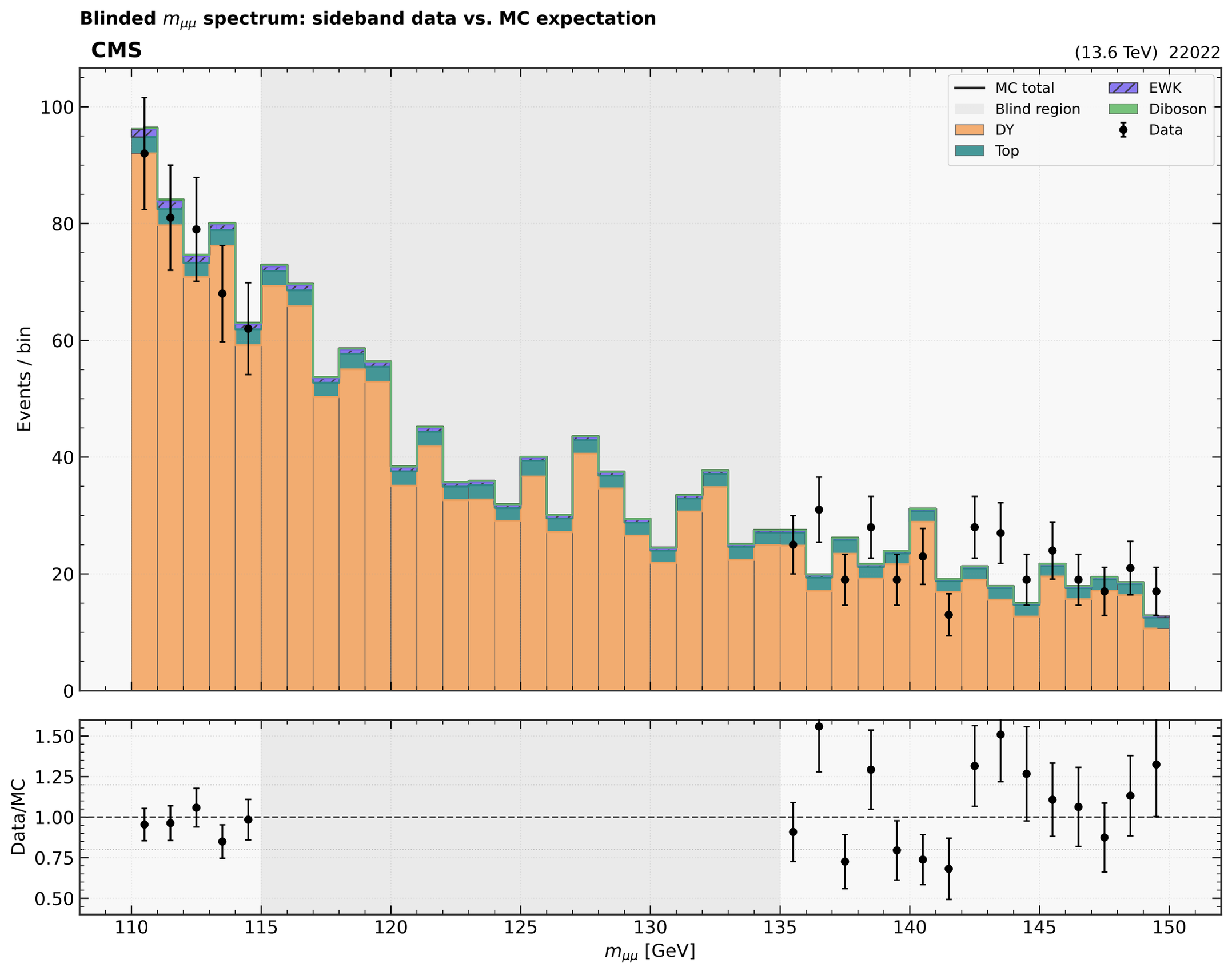


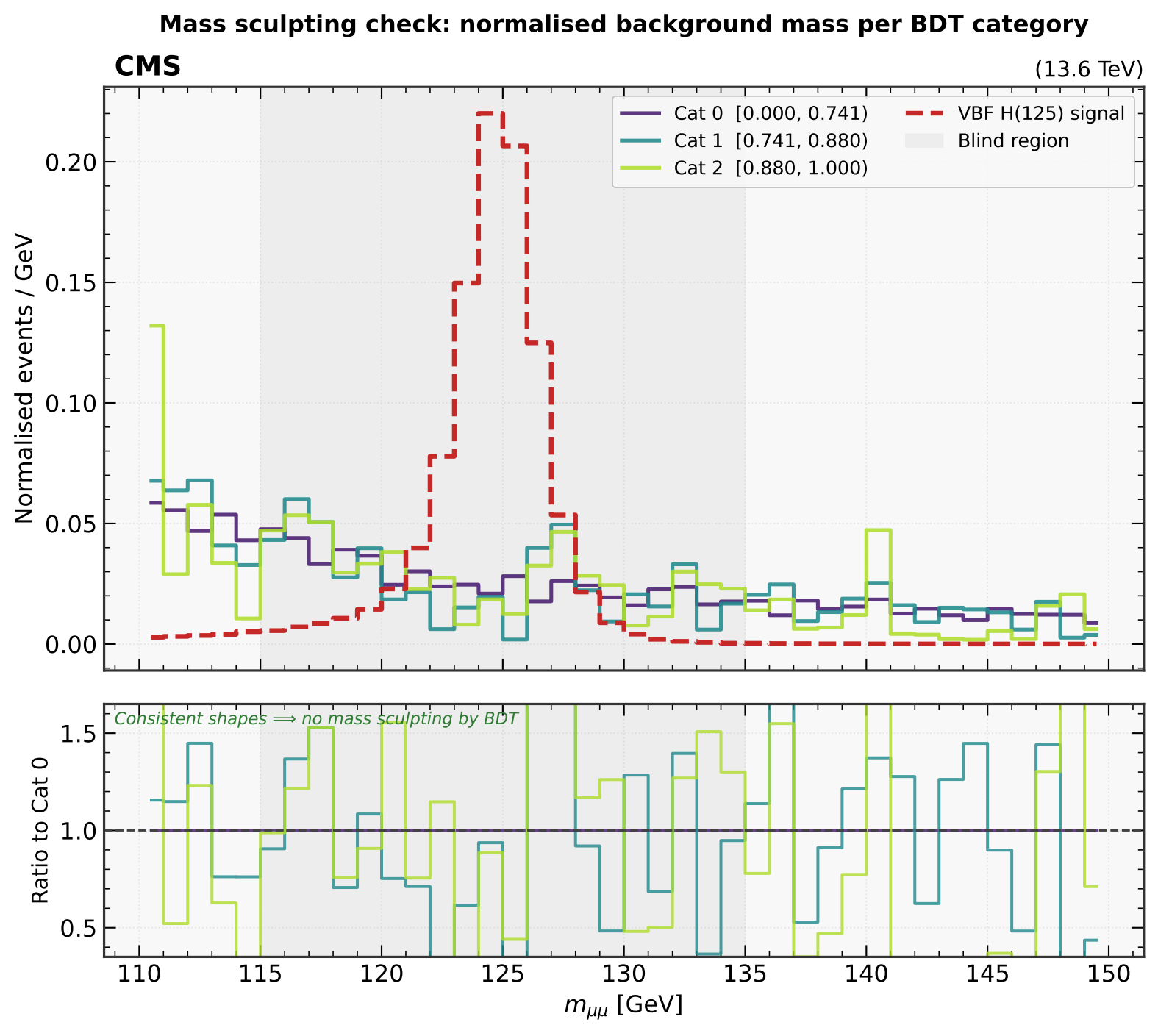
VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window



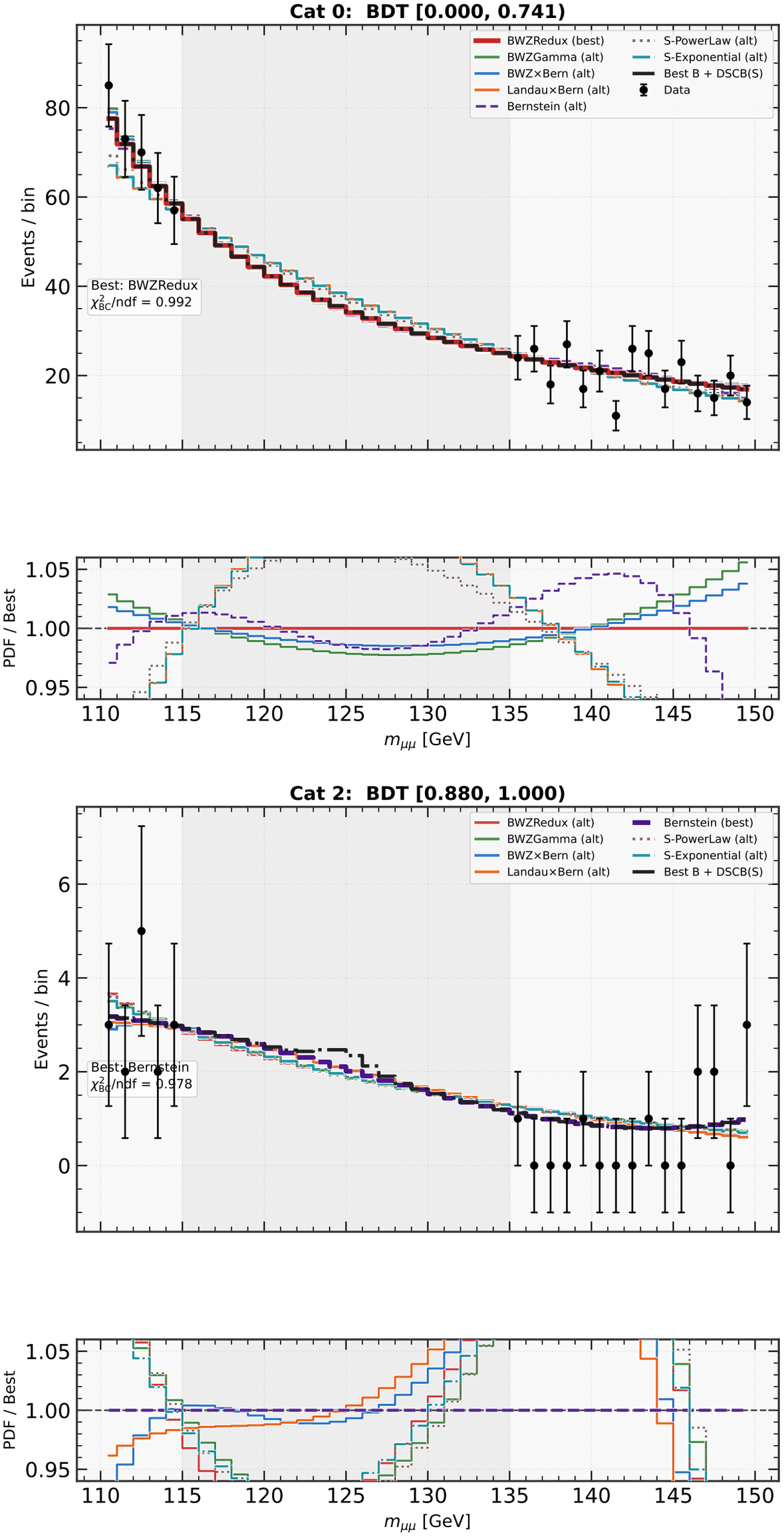

Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 3.84 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
3.4
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

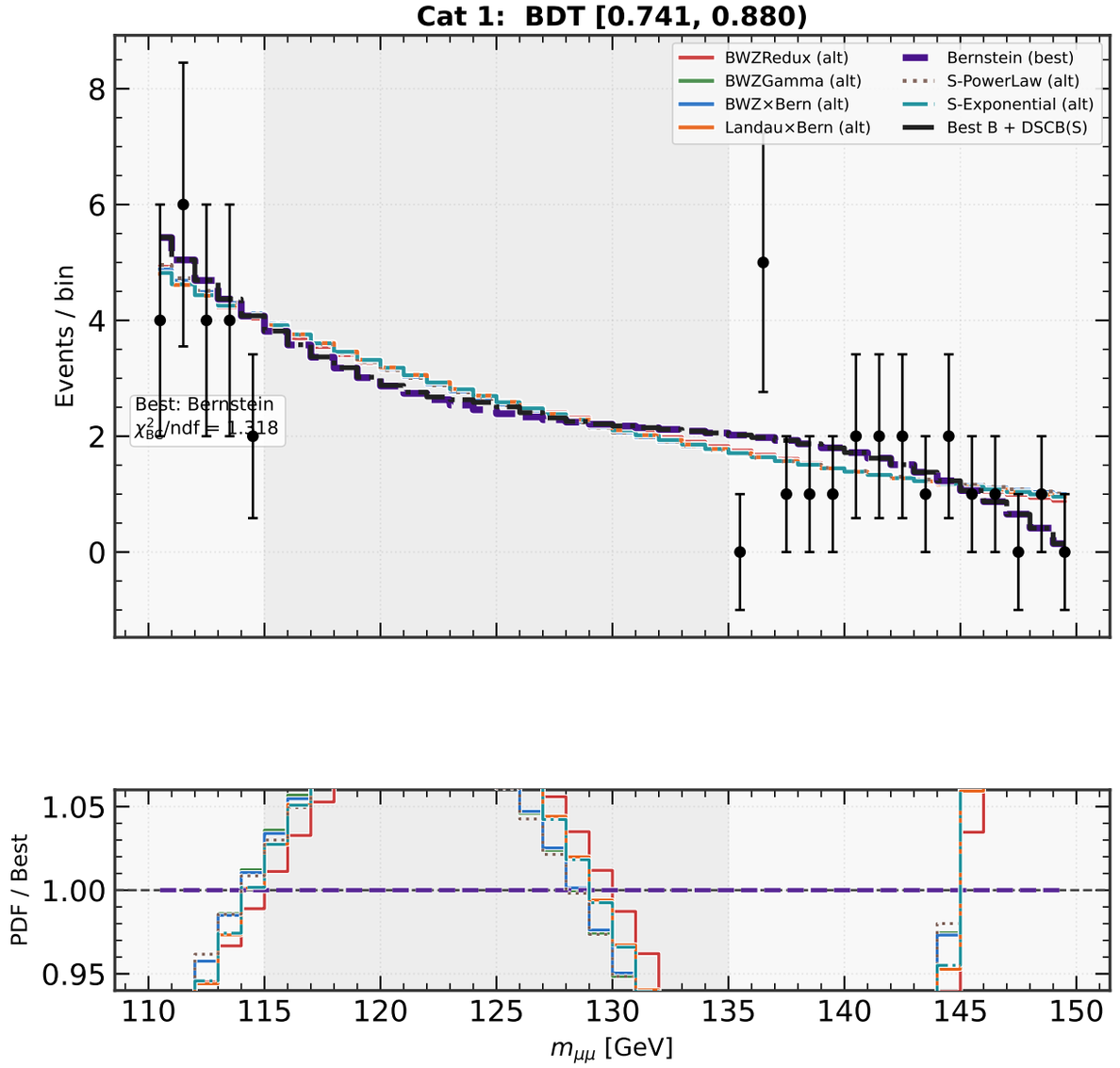

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 22022 (AUC = 0.888)


Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 22022
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]


Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.741)
Cat 1 [0.741, 0.880)
Cat 2 [0.880, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.00
0.05
0.10
0.15
0.20
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
0.5
1.0
1.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits

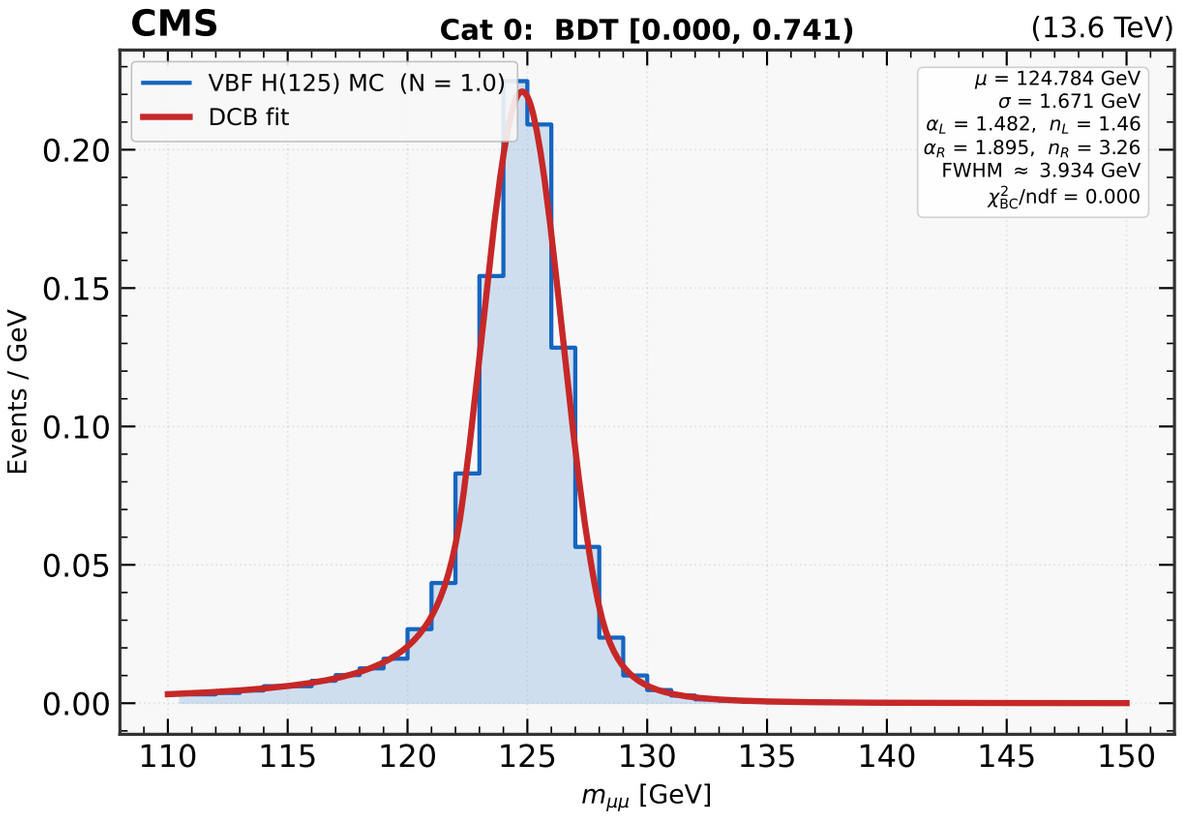


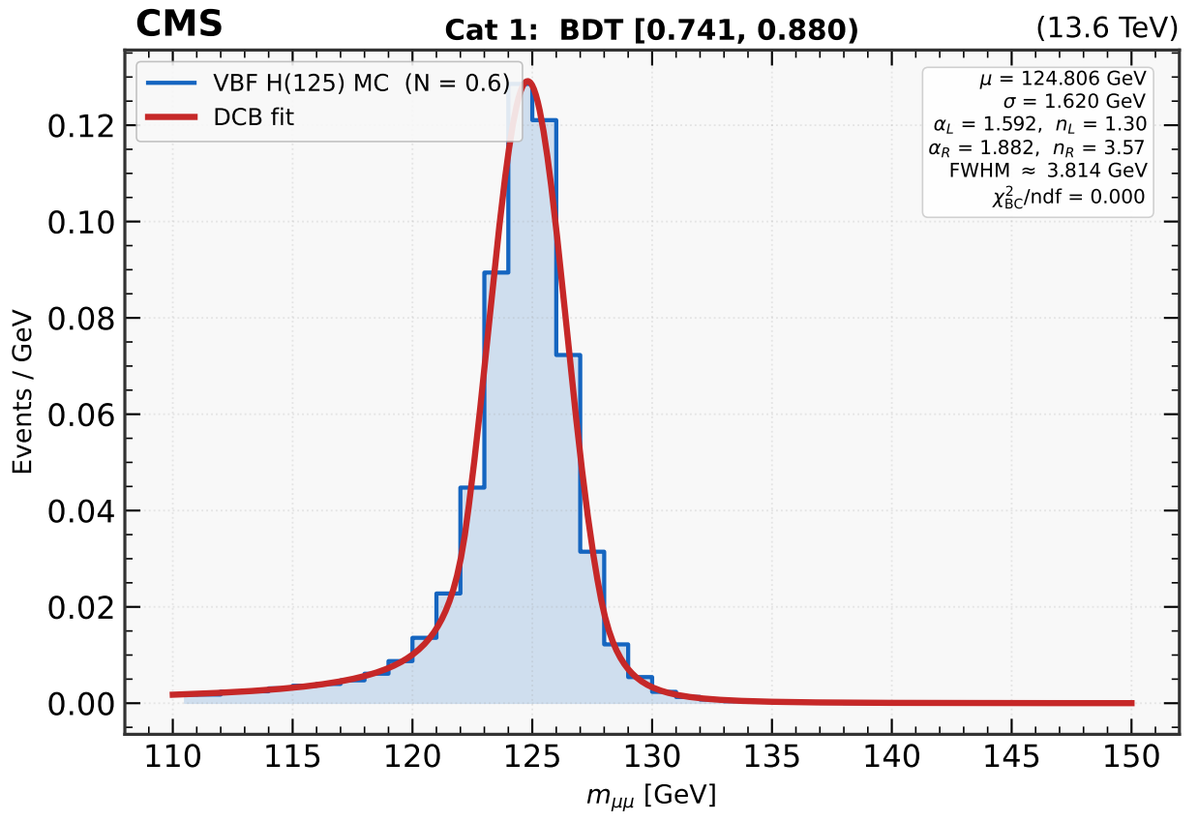


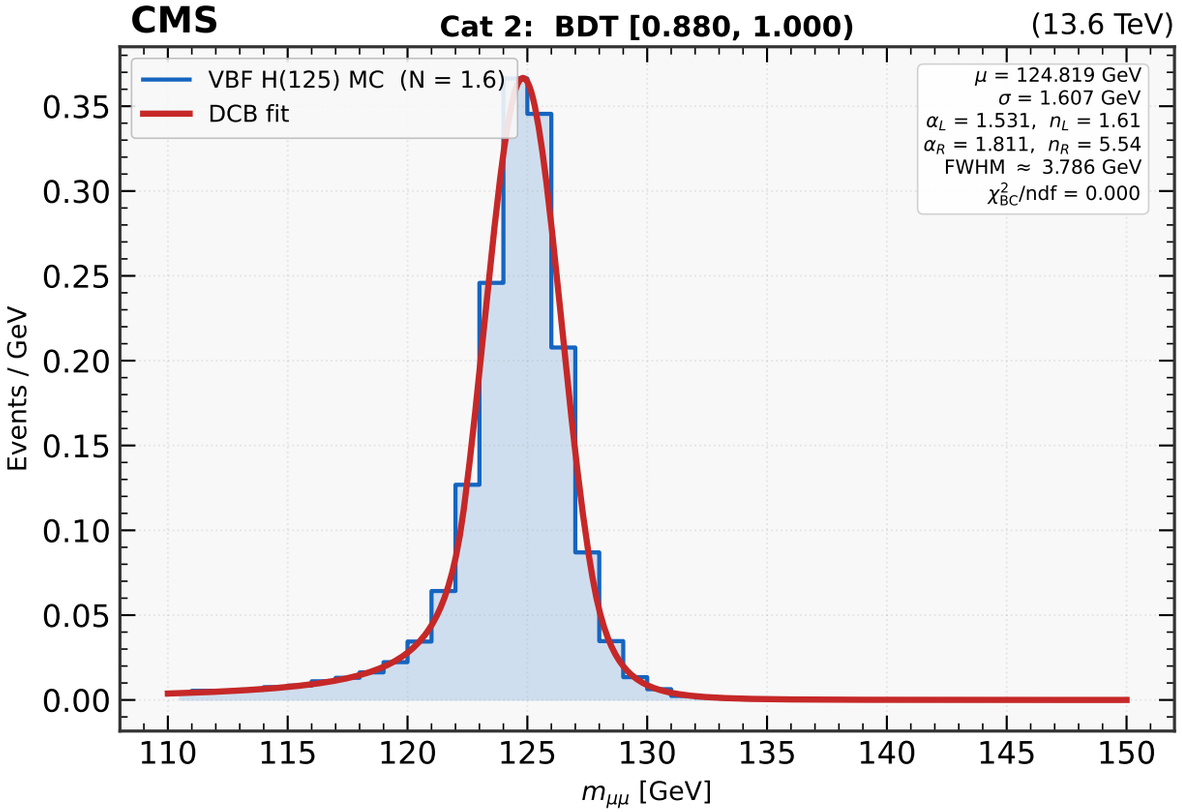
Per-category background model fit comparison (χ²_BC/ndf, ndf)

Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.741)	BWZRedux	0.992	0.992 (37)	1.006 (37)	1.027 (36)	1.261 (37)	1.098 (35)	1.160 (37)	1.248 (37)
1	[0.741, 0.880)	Bernstein	1.318	1.320 (37)	1.328 (37)	1.364 (36)	1.326 (37)	1.318 (35)	1.327 (37)	1.326 (37)
2	[0.880, 1.000)	Bernstein	0.978	1.041 (37)	1.024 (37)	1.044 (36)	1.020 (37)	0.978 (35)	1.027 (37)	1.026 (37)

VBF H(125) signal DCB extraction per BDT category

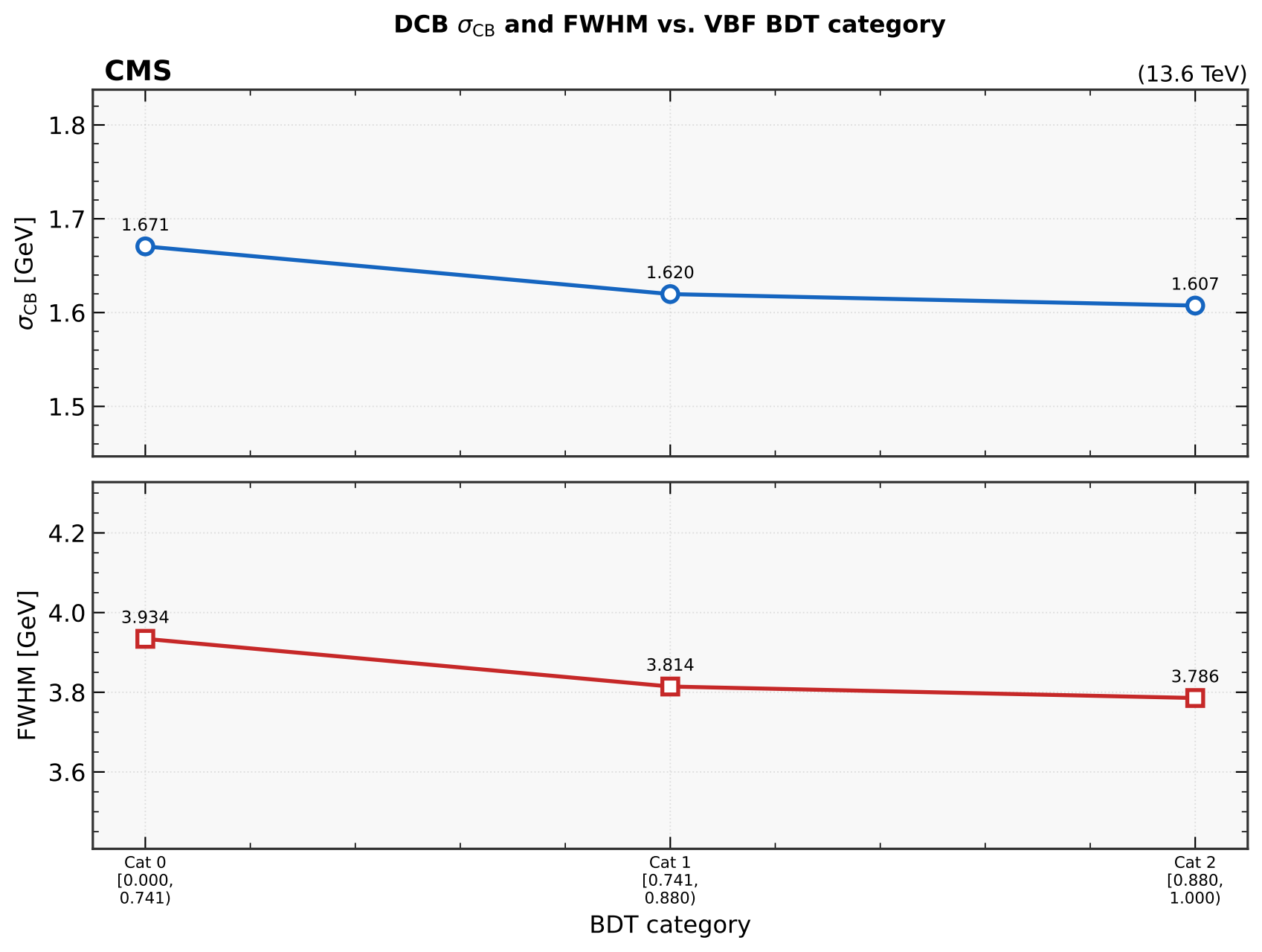


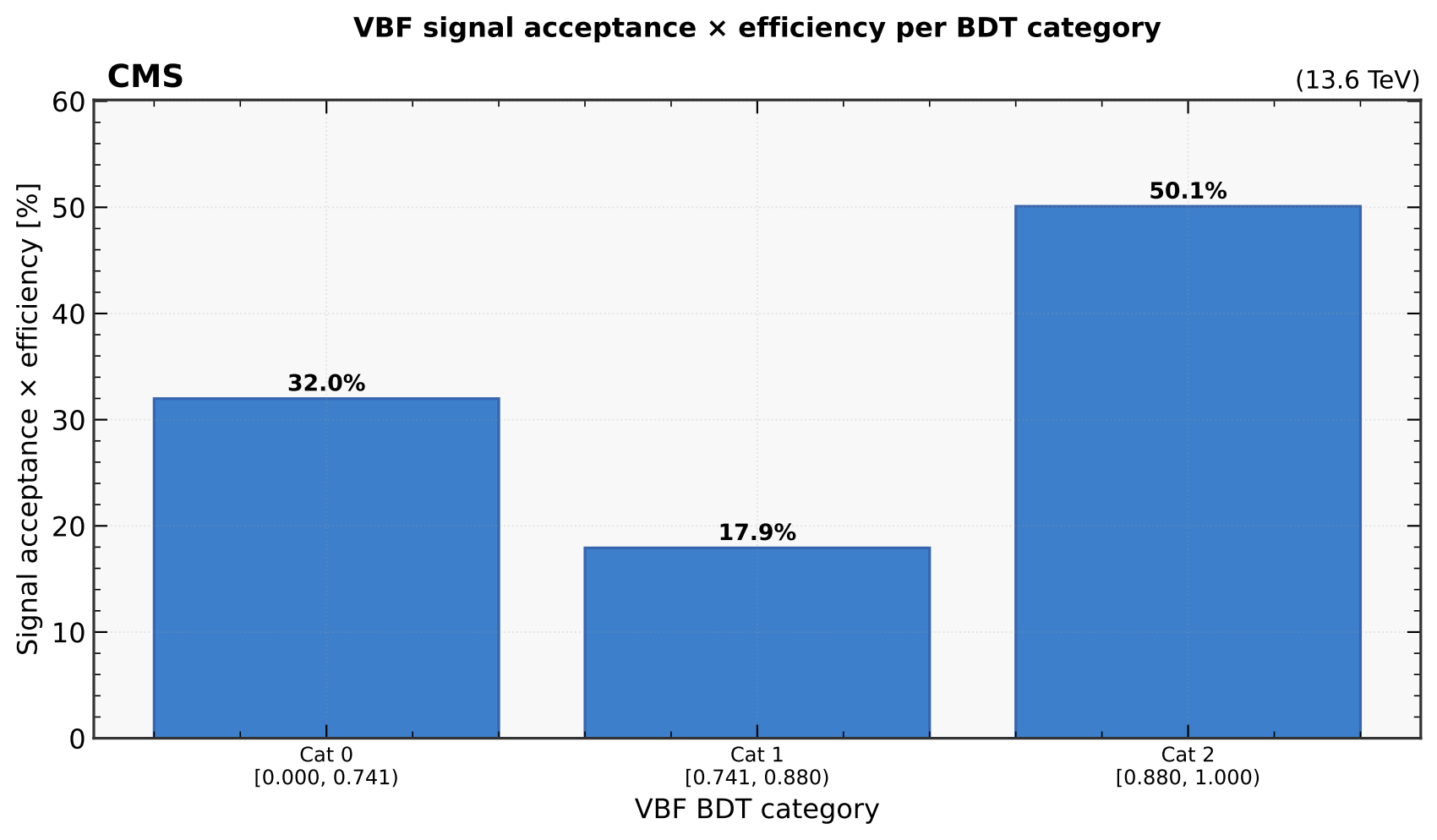




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.741)	1.05	124.7841	1.6705	1.4817	1.461	1.8950	3.262	3.934	0.040	0.000
1	[0.741, 0.880)	0.59	124.8056	1.6197	1.5921	1.301	1.8820	3.568	3.814	0.105	0.000
2	[0.880, 1.000)	1.64	124.8191	1.6075	1.5311	1.607	1.8108	5.535	3.786	0.351	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.8
1.7
1.6
1.5
1.671
1.620
1.607
FWHM [GeV]
4.2
4.0
3.8
3.6
3.934
3.814
3.786
Cat 0
[0.000,
0.741)
Cat 1
[0.741,
0.880)
Cat 2
[0.880,
1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
60
50
40
30
20
10
0
32.0%
17.9%
50.1%
Cat 0
[0.000, 0.741)
Cat 1
[0.741, 0.880)
Cat 2
[0.880, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.741)	1.05	3498.48	0.713	319.676	[122.8, 126.8]	0.040	0.040
1	[0.741, 0.880)	0.59	200.93	0.400	14.515	[122.9, 126.7]	0.105	0.105
2	[0.880, 1.000)	1.64	190.69	1.133	10.074	[122.9, 126.7]	0.357	0.351

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.741)	647	1620.60 ± 43.65	0.399	3498.48 ± 64.26	0.0399	0.0399	319.676
1	[0.741, 0.880)	40	97.03 ± 10.39	0.412	200.93 ± 15.03	0.1051	0.1046	14.515
2	[0.880, 1.000)	25	81.97 ± 9.64	0.305	190.69 ± 14.96	0.3569	0.3505	10.074
Combined	all						0.3680	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total_err2	total
0	[0.000, 0.741)	3243.91	26.71	216.68	11.18	4129.37	3498.48
1	[0.741, 0.880)	183.52	8.76	7.79	0.86	226.01	200.93
2	[0.880, 1.000)	165.62	19.18	5.32	0.57	223.82	190.69